

SAMPRIT MANDAL

📍 Kolkata, India 📞 +91-8910856627 ✉ Samprit.code@gmail.com
🌐 sampritdev.vercel.app 🐙 [Github](#) 🔗 [LinkedIn](#)

PROFESSIONAL SUMMARY

3rd-year Computer Science and Engineering (AI and ML) student with a primary focus on Machine Learning and Deep Learning, proven expertise as a Full Stack Developer and Web3 Enthusiast. Skilled in architecting modern web applications using React, Node.js, and cutting-edge technologies. Demonstrated ability to craft responsive, high-performance UIs and scalable backends while delivering impactful digital experiences through clean code and elegant design. Active open-source contributor with hands-on experience across multiple production environments.

PROFESSIONAL EXPERIENCE

Frontend Developer

Etailled Digital Services Private Limited

Oct 2025 – Mar 2026

Remote

- Engineered responsive UI components using React and modern CSS frameworks, improving user engagement by 40%
- Collaborated with design teams to implement pixel-perfect interfaces and optimize web performance by 35%

FEATURED PROJECTS

Smart AI Study Assistant (Multi-Agent LLM System with RAG)

[GitHub](#)

- Developed a multi-agent LLM study assistant generating personalized study plans, quizzes, and flashcards | **Tech:** Python, Streamlit, Google Gemini, ChromaDB
- Built a Retrieval-Augmented Generation (RAG) pipeline with ChromaDB vector store and hybrid vector + keyword search
- Implemented SM-2 spaced repetition progress tracking, gamification, study scheduling, and multi-language UI (7 languages)

Glowify – AI-Powered Cosmetic Recommendation Platform

[GitHub](#)

- Created intelligent recommendation engine using machine learning to suggest personalized cosmetic products | **Tech Stack:** React, Python, Flask, Node.js, TensorFlow, Scikit-learn, MongoDB
- Built custom ML model with 85%+ accuracy for skin type classification and product matching
- Integrated marketplace features with user reviews, ratings, and secure checkout functionality

VisuLearn AI – Visual Learning Platform for DSA and ML

[GitHub](#) | [Open Source](#)

- Developed open-source educational platform with interactive visualizations for Data Structures, Algorithms, and ML concepts
- Tech Stack:** React, D3.js, Python, FastAPI, WebGL
- Created 50+ algorithm visualizations with step-by-step execution and complexity analysis
- Executed real-time code editor with syntax highlighting and algorithm performance comparison

Plant Disease Detection – QCNN with HSI

[GitHub](#) | [Research](#)

- Developed novel Quantum Convolutional Neural Network (QCNN) with Squeeze-and-Excitation blocks for plant disease detection
- Tech Stack:** Python, PyTorch, Qiskit, OpenCV, Hyperspectral Imaging
- Achieved 92% accuracy on multi-class disease classification using hyperspectral imaging data
- Established attention mechanisms to identify stress patterns invisible to traditional RGB imaging

kickResume – Smart Resume Builder

[GitHub](#)

- Built full-stack resume builder with 10+ customizable templates, real-time preview, and one-click PDF export functionality
- Tech Stack:** React, Node.js, TypeScript, Express, MongoDB, JWT Authentication
- Established drag-and-drop interface and auto-save feature, improving user retention by 40%
- Integrated ATS-friendly parsing algorithms to optimize resume content for applicant tracking systems

Manzil – Group Tour and Ride Navigation App

[GitHub](#)

- Designed a real-time group navigation app with live location tracking, trip coordination, SOS alerts, and member navigation

- Architected real-time communication with background geolocation, push notifications, authentication, and geospatial tracking

EDUCATION

Bachelor of Technology in Computer Science and Engineering (AI and ML)

2023 – 2027

Techno Bengal Institute of Technology

Kolkata, India

- Relevant Coursework: Data Structures and Algorithms, Machine Learning, Deep Learning, Web Development, Database Management, Computer Networks, Operating Systems, Software Engineering

TECHNICAL SKILLS

- **Languages:** C++, Python, JavaScript, TypeScript, SQL
- **Frontend:** React.js, Next.js, Redux, Tailwind CSS, D3.js
- **Backend:** Node.js, Express.js, Flask, FastAPI, REST APIs
- **Databases:** MongoDB, PostgreSQL, MySQL, Firebase
- **AI/ML:** PyTorch, TensorFlow, OpenCV, RAG, Computer Vision, Scikit-learn
- **Tools & Cloud:** Git, GitHub, Docker, AWS, CI/CD

ADDITIONAL INFORMATION

- **Current Focus:** Actively working on multiple projects while continuously upskilling in Web and App Development, AI/ML model optimization, and Web3 technologies (DApps, Smart Contracts, Blockchain Infrastructure)
- **Languages:** English (Professional), Hindi (Professional), Bengali (Native)
- **Interests:** Open-source contribution, Quantum Computing, Artificial Intelligence and Machine Learning, Blockchain technology, UI/UX Design